

TUNISIAN ENGINEERING CONGRESS WEEK

# TECWEEK

IGNIS 3.0

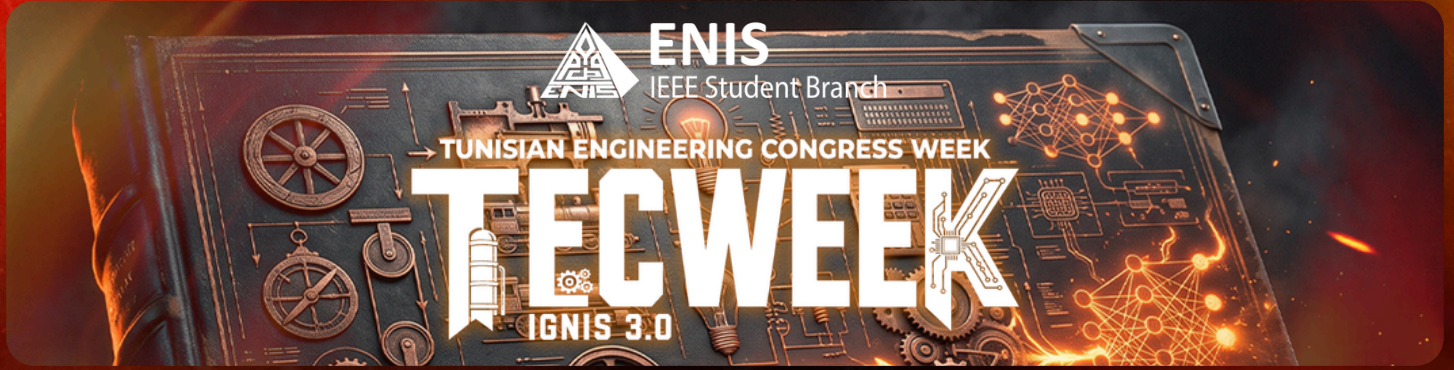
## TECHNICAL CHALLENGE

## SPECIFICATION BOOK



ENIS

IEEE Student Branch



## OBSERVE & ACT: INTELLIGENT MONITORING SYSTEMS

**THEME: ENGINEERING BEYOND LIMITS — INVENT. BUILD. TRANSFORM**

### CHALLENGE OVERVIEW & VISION:

Throughout history, engineering has continuously evolved, moving from purely mechanical systems toward advanced technologies capable of sensing, analyzing, and supporting autonomous decisions. Today, as the world faces growing climate, industrial, urban, and societal challenges, the way we monitor and interact with our environment must also evolve. Engineering is entering a new era where innovation is no longer limited to observation and measurement. Modern systems are transforming from passive monitoring tools into intelligent solutions capable of understanding situations, generating insights, and supporting real-time actions. Traditional monitoring systems mainly focus on collecting and recording information. However, data alone is not enough to address complex real-world problems. The future of engineering requires systems that can transform observations into meaningful decisions and responsible actions. Modern engineering demands systems that do not merely report problems, but actively solve them. The “Observe & Act” challenge invites multidisciplinary teams to design intelligent monitoring solutions that connect digital intelligence with real-world impact. Participants are encouraged to create systems that follow a complete process:

**Detect** ➔ **Understand** ➔ **Alert** ➔ **Act**

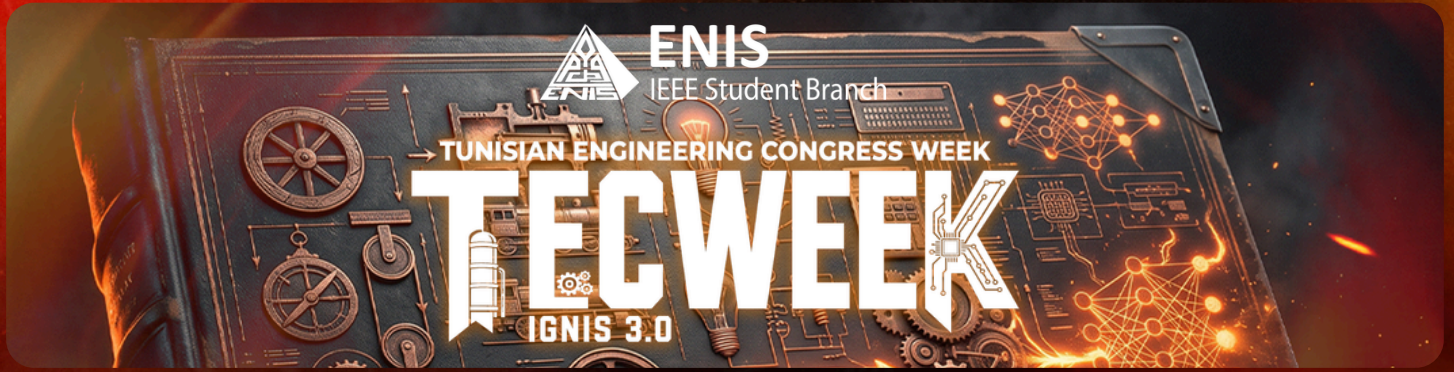
Through this challenge, we explore how engineering can combine innovation, intelligence, and responsibility to develop solutions that serve humanity and contribute to a smarter, safer, and more sustainable future.

### CHALLENGE OBJECTIVE

Teams must identify a real-world problem and propose an intelligent monitoring solution capable of not only detecting relevant information from its environment, but also understanding and analyzing the collected information, identifying patterns or potential risks, making appropriate decisions based on the situation, and generating alerts, recommendations, or automated actions.

### APPLICATION FIELDS

Teams are encouraged to develop intelligent monitoring solutions in various domains. Examples include: Energy & Smart Infrastructure: Energy optimization, smart grids, smart buildings. Environment & Natural Resources: Pollution monitoring, waste management, water quality monitoring. Smart Campus & Human Well-being: Campus safety, smart parking, facility monitoring. Agriculture & Rural Solutions: Smart irrigation, crop monitoring, livestock tracking. Industry & Manufacturing: Predictive maintenance, machine monitoring, industrial automation. Health, Well-being & Sports: Health monitoring, wearable devices, performance tracking. Community & Smart Services: Disaster alerts, food waste reduction, smart retail, cybersecurity monitoring. Teams may also explore other fields as long as their solution follows the workflow



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#### Alignment with the United Nations Sustainable Development Goals (SDGs)

Each project should contribute to at least one of the 17 United Nations Sustainable Development Goals (SDGs). Teams are expected to clearly explain:

- **Target SDG(s):** Identify the primary SDG(s) your project supports and briefly justify your choice.
- **Proposed Impact:** Describe how your solution addresses the identified challenge and the expected benefits for users or the community.
- **Impact Indicators:** Provide measurable outcomes or success metrics (when applicable) to demonstrate the potential effectiveness of your solution.

#### METHODOLOGICAL approach

To help teams develop their solutions in a structured and engineering-oriented manner, TECWEEK encourages participants to follow the 6-stage Project Development Lifecycle:

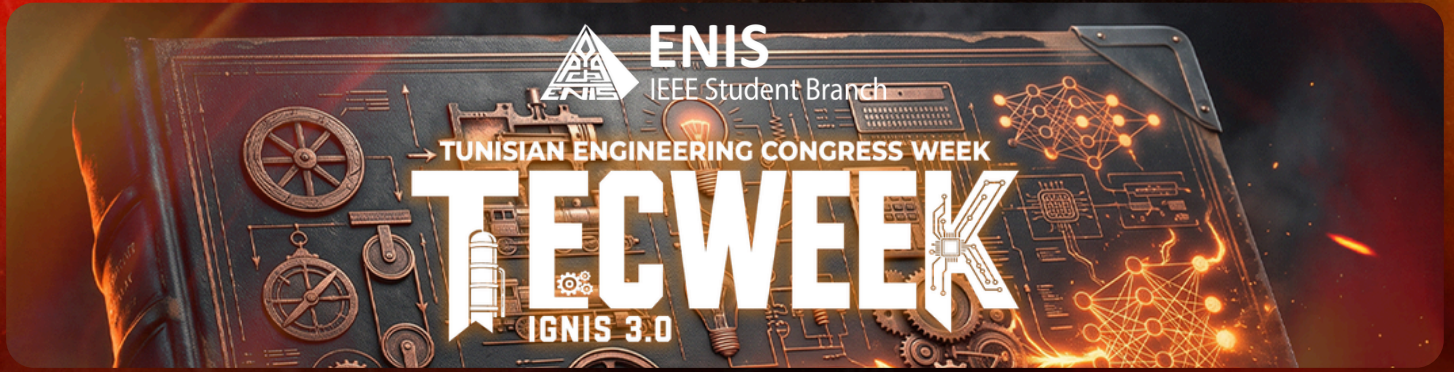
- **EXPLORE:** Research the problem domain, analyze existing solutions, investigate user needs, and collect relevant data.
- **DEFINE :** Formulate a clear problem statement, identify the target users, define the project scope, and align the solution with one or more Sustainable Development Goals (SDGs).
- **DESIGN :** Design the proposed solution by defining the system architecture, selecting the required technologies, and illustrating the workflow
- **BUILD :** Develop a functional prototype, whether software, hardware, or a hybrid system, and integrate the selected technologies.

- **VALIDATE :** Evaluate the solution through testing, collect user or stakeholder feedback when possible, and assess its technical performance and expected real-world impact.
- **PITCH :** Present the problem, solution, prototype, technical approach, feasibility, and impact in a clear and convincing final presentation to the jury.

#### EVALUATION RUBRIC

Projects are evaluated out of 100 base points, with up to +15 bonus points available:

| Evaluation Category                             | Description & Jury Expectations                                                                                                                                               | Points         |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Innovation &amp; Originality</b>             | Is the idea novel? Does it show a clear evolution compared to traditional passive methods?                                                                                    | <b>25 pts</b>  |
| <b>Problem Definition &amp; SDG Impact</b>      | Is the problem real, targeted, and backed by a real stat/fact? Are target users and SDG impact well-identified?                                                               | <b>20 pts</b>  |
| <b>System Logic &amp; Architecture</b>          | Evaluation of the system design, workflow, technology choices, AI models or decision logic used, and the coherence between data collection, analysis, decisions, and actions. | <b>25 pts</b>  |
| <b>Feasibility &amp; Realism</b>                | Real-world impact, technical feasibility, and sustainability.                                                                                                                 | <b>20 pts</b>  |
| <b>Presentation &amp; Communication</b>         | Cumulative evaluation of the Innovation Brief, Final Pitch Deck, and Oral Presentation.                                                                                       | <b>10 pts</b>  |
| <b>BASE SCORE MAXIMUM</b>                       | Cumulative evaluation of Innovation Brief and Final Pitch Deck                                                                                                                | <b>100 pts</b> |
| <b>BONUS – Prototype or live demo presented</b> | Software app/UI, Hardware circuit, Hybrid system, or Low-tech physical model                                                                                                  | <b>+15 pts</b> |
| <b>MAXIMUM POSSIBLE SCORE</b>                   | Total potential score including all bonus awards                                                                                                                              | <b>115 pts</b> |



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#### DELIVERABLES & SUBMISSION REQUIREMENTS

All deliverables must be submitted via the official platform before the deadlines.

- **Phase 1: Ideation & Innovation Brief**
  - **Concise Presentation (required)**
    - Your slides should highlight the main idea, design choices, and results, covering the problem, solution, results, roadmap, challenges, and team.
  - **Technical Report (required)**
    - Clear and simple (3–5 pages).
    - Must include:
      - Problem tackled & why it matters.
      - Algorithms or decision logic used.
      - Testing results (simulation screenshots or prototype photos, if you built a prototype).
    - Mention:
      - **Value Proposition:** What you sell and why people would choose your solution over existing alternatives.
      - **Target Market:** The exact group of people or businesses you plan to serve.
      - **Revenue Streams:** How your company earns money (such as sales, ads, or service fees).
      - **Cost Structure:** The major expenses required to keep the business running.
      - All statistics, facts, and claims must be supported by reliable, traceable sources.

No elimination in Phase 1 every participating team proceeds to the next stage.

- **Phase 2: Grand Finale Pitch**
  - **Pitch Presentation:** 5 minutes (including an optional 2–3 minute video showcasing the system's functionality).
  - **Jury Q&A Session:** 3 minutes.
  - **Prototype (optional, encouraged):** Either a physical prototype or a software simulation.

#### REFERENCES (Recommended Resources)

Participants are encouraged to consult the following resources:

- **IEEE Xplore Digital Library** – Research papers on Artificial Intelligence, IoT, Embedded Systems, Smart Monitoring, Computer Vision, and Sustainable Technologies.
- **Google AI** – Beginner resources on Machine Learning and Artificial Intelligence.
- **Microsoft Learn** – AI, IoT, and cloud learning paths.
- **Arduino, ESP32, and Raspberry Pi** Official Documentation – For hardware-based projects.
- **UN Sustainable Development Goals (SDGs)** – Inspiration for societal impact and sustainability.
- Recent publications, reports, and datasets related to your chosen problem domain.

#### ELIGIBILITY & RULES

1. **Eligibility:** Open to university / higher education students across all academic disciplines and experience levels.
2. **Team Size:** Teams of 2 to 4 members. (ieee or non ieee members).
3. **Languages:** Submissions and presentations must be entirely in English or entirely in French.

#### IMPORTANT DATES

- **Mentorship sessions** will be announced soon
- **Phase 1 Submission Deadline:** September 24, 2026 at 23:59 (GMT+1)
- **Phase 2 Grand Finale Pitch:** September 27, 2026

#### CONTACT & LINK

- **Registration Link:**  
<https://forms.gle/vYyzzna6k3MTjh476>
- **Organization Contact:**
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